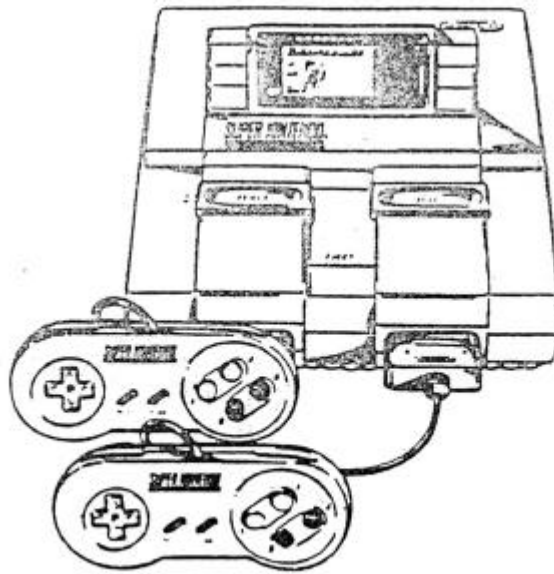


SUPER NINTENDO. **ENTERTAINMENT SYSTEM**



TECHNICAL MANUAL

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SUPER NINTENDO

ENTERTAINMENT SYSTEM

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English Translation — Fully Corrected & Completed

Based on the incomplete Playtronic Brazilian Portuguese service manual translation originally started by Mark (TheRetroChannel). All residual Portuguese terminology, grammar issues, and technical inaccuracies have been corrected.

July 2026

INTRODUCTION

This is a cleaned and completed English translation of the Playtronic Super NES service manual (Manual Técnico). The original manual was written in Brazilian Portuguese around 1994 and primarily covers the SHVC revision of the console, with a brief mention of the later GPM (Ground Plane Method) revision.

Due to the lack of widely available official English service documentation for the Super NES, an incomplete translation was previously released. That version was estimated to be approximately 50 % complete and contained residual Portuguese terms, awkward phrasing, and OCR-related errors. The present document corrects those issues and completes the remaining sections (Controller, Super Scope, Mouse, Cartridge, IC handling notes, and Glossary).

All valuable service information for the console itself, including the burn-in test cartridge fault priority table, disassembly procedures, and diagnostic tables, has been retained and polished. Text that appeared only inside scanned images in the original (and was therefore untranslated) has been reconstructed from context and standard Nintendo terminology where possible.

Important: This manual is intended for experienced technicians. The Super NES contains static-sensitive components. Always observe proper ESD precautions when handling the main board or ICs.

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1. OVERVIEW

Console

The Super NES console uses a Central Processing Unit (CPU), a two-IC Picture Processing Unit (PPU1 & PPU2), and two sound microprocessors (S-SMP and S-DSP).

1.1 CPU — Central Processing Unit

- Custom Nintendo processor (Ricoh 5A22, based on the 65C816)
- 24-bit address bus allowing direct access to 16 MB of ROM/RAM address space
- 16-bit internal registers
- 8-bit system data bus
- Built-in Direct Memory Access (DMA) and HDMA controllers

1.2 PPU1 / PPU2 — Picture Processing Units

- Custom Nintendo processors (two ICs)
- Up to 128 sprites on screen, each with independent priority; maximum 32 sprites per scanline
- Sprite sizes: 8×8, 16×16, 32×32, or 64×64 pixels
- 32,768 colors available (15-bit RGB)
- Eight 16-color palettes available for backgrounds and sprites
- Horizontal, vertical, and diagonal scrolling; partial vertical scrolling supported
- Resolutions: 256×224, 512×224, or 512×448 (interlaced)
- Special effects: mosaic, windowing, fixed-color add/subtract, screen add/subtract, pseudo-hires
- 16-bit data bus to VRAM

1.3 Sound Microprocessors

- Eight channels of 16-bit Pulse Code Modulation (PCM) sound
- Stereo digital audio output
- Dedicated S-SMP (Sony SPC700-based) and S-DSP chips with 64 KB shared ARAM

An AC adapter is supplied with the Super NES that converts 120 V AC to approximately 10 V unregulated DC.

Controller Devices

Controller devices allow the player to interface with the game. The standard controller has twelve input functions. The CPU periodically loads parallel button data into serial shift registers and reads the serial bitstream. This process occurs at software-controlled intervals, typically every frame (~16.7 ms NTSC).

Super Scope

The Super Scope is a peripheral light-gun device consisting of a transmitter (gun) and a receiver assembly. The transmitter detects the timing of the CRT electron beam to determine the aim point. This information is transmitted via infrared to the receiver, which interfaces with the console.

Game Cartridge

Game cartridges contain semiconductor memory ICs in various combinations of program ROM and work RAM. A CIC (Checking Integrated Circuit / lockout chip) communicates with a matching CIC in the console to prevent unauthorized software from running. Some cartridges also include battery-backed SRAM for save data, DSP-1, or Super FX co-processors.

2. SUPER NES CONSOLE OPERATION

Purpose

The Super NES Console contains the system hardware. Its purpose is to execute the program stored in the game cartridge ROM, generate video graphics via the PPUs, produce sound, and accept player input through controller ports. The console can be connected to a television or monitor. The multi-out connector provides stereo AV, RGB, and S-Video. A separate RF modulator provides monaural RF output via an RF switch. A 28-pin expansion port allows connection of external peripherals.

Theory of Operation

The console cannot operate by itself; it requires the program ROM on a game cartridge and (usually) player input. All activity is coordinated by the Central Processing Unit (CPU).

The console can be subdivided into eight functional blocks:

1. Power Supply Section
2. Controller Input Section
3. System Timing Section
4. RESET Section
5. RESET IC Section (voltage supervisor)
6. Sound Section
7. Display Section
8. Program Section (CPU + Work RAM)

2.1 Power Supply Section

The AC adapter supplies ~10 V DC unregulated. On the main PCB this is filtered, protected against reverse polarity and surges, switched, and regulated to +5 V (Vcc) for logic and ~7 V unregulated (Vs) for the audio amplifier.

2.2 Controller Input Section

Includes diode arrays and capacitors for ESD/spike protection plus pull-up resistors. The CPU reads controller data approximately once per frame, although software can vary the timing.

2.3 System Timing Section

A 21.47727 MHz (NTSC) crystal oscillator generates the master clock distributed to the CPU, both PPUs, and the cartridge connector. The CPU divides this clock to produce internal operating frequencies (approximately 3.58 MHz, 2.68 MHz, and 1.79 MHz). DMA always runs at the slower rate. Access speed automatically adjusts according to the memory region being addressed.

2.4 RESET Section

Uses a 4 MHz ceramic resonator, a 74HCU04 inverter, the CIC security chip, the reset button, and associated discrete components. The console CIC communicates with the cartridge CIC. To ensure correct power-on sequencing, the CIC reset is applied first to PPU2 (which introduces delay), then to PPU1, and finally to the CPU.

2.5 RESET IC Section

A voltage detector IC (PST529 or equivalent) holds the system in reset during power-up, power-down, or brown-out conditions. This prevents corrupted data from being written to battery-backed cartridge RAM.

2.6 Sound Section

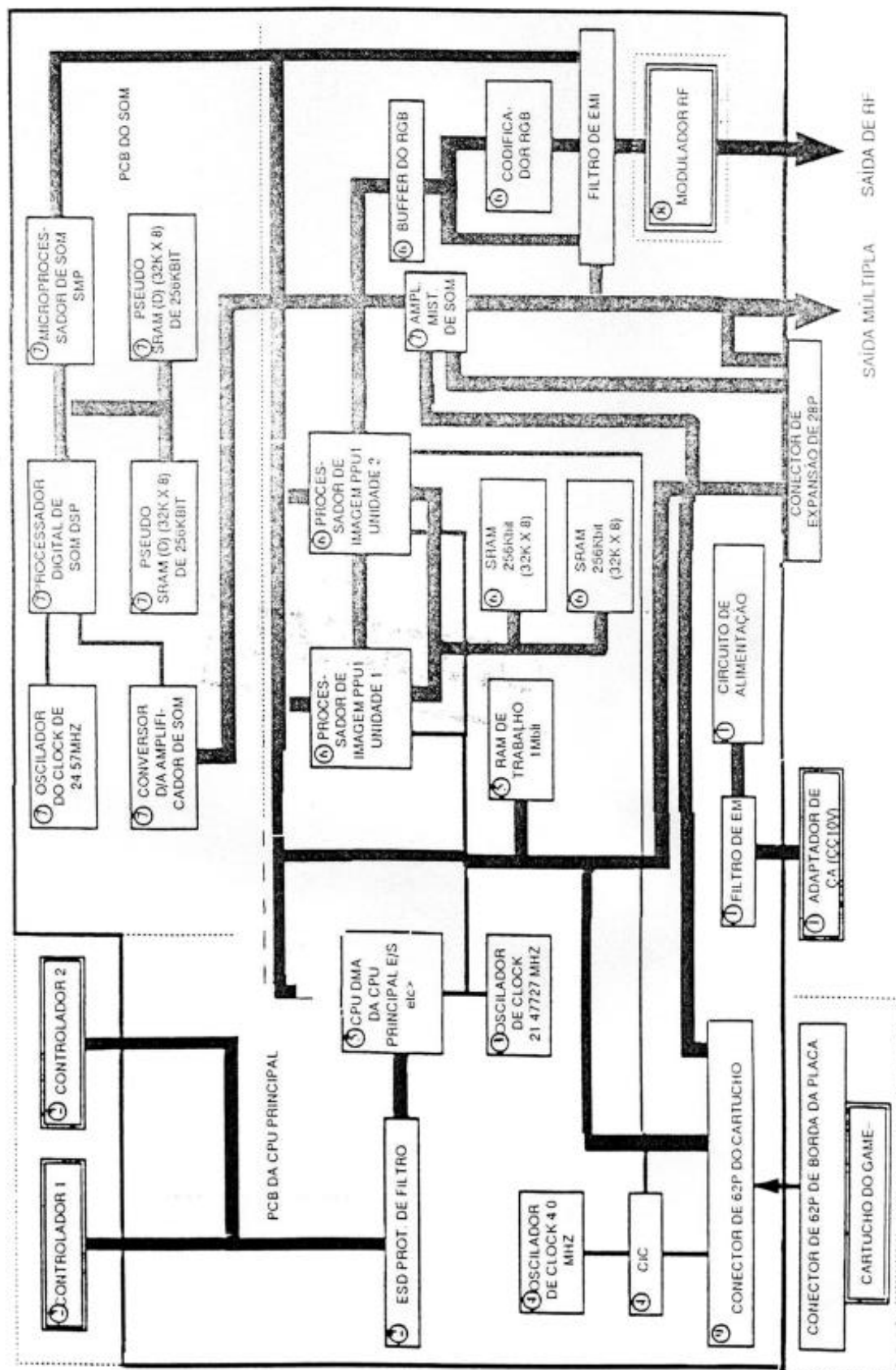
Consists of a plug-in sound module (or integrated circuitry on later boards) containing the S-SMP, S-DSP, DAC, and two 32 KB SRAM chips (64 KB total ARAM). Program and sample data are transferred from cartridge ROM into ARAM under SMP control. The DSP synthesizes eight PCM channels with noise, echo, envelope, and pitch modulation. Stereo output is amplified by an LM358 (or equivalent) op-amp pair and routed to the multi-out connector. RF and expansion audio are mixed to mono.

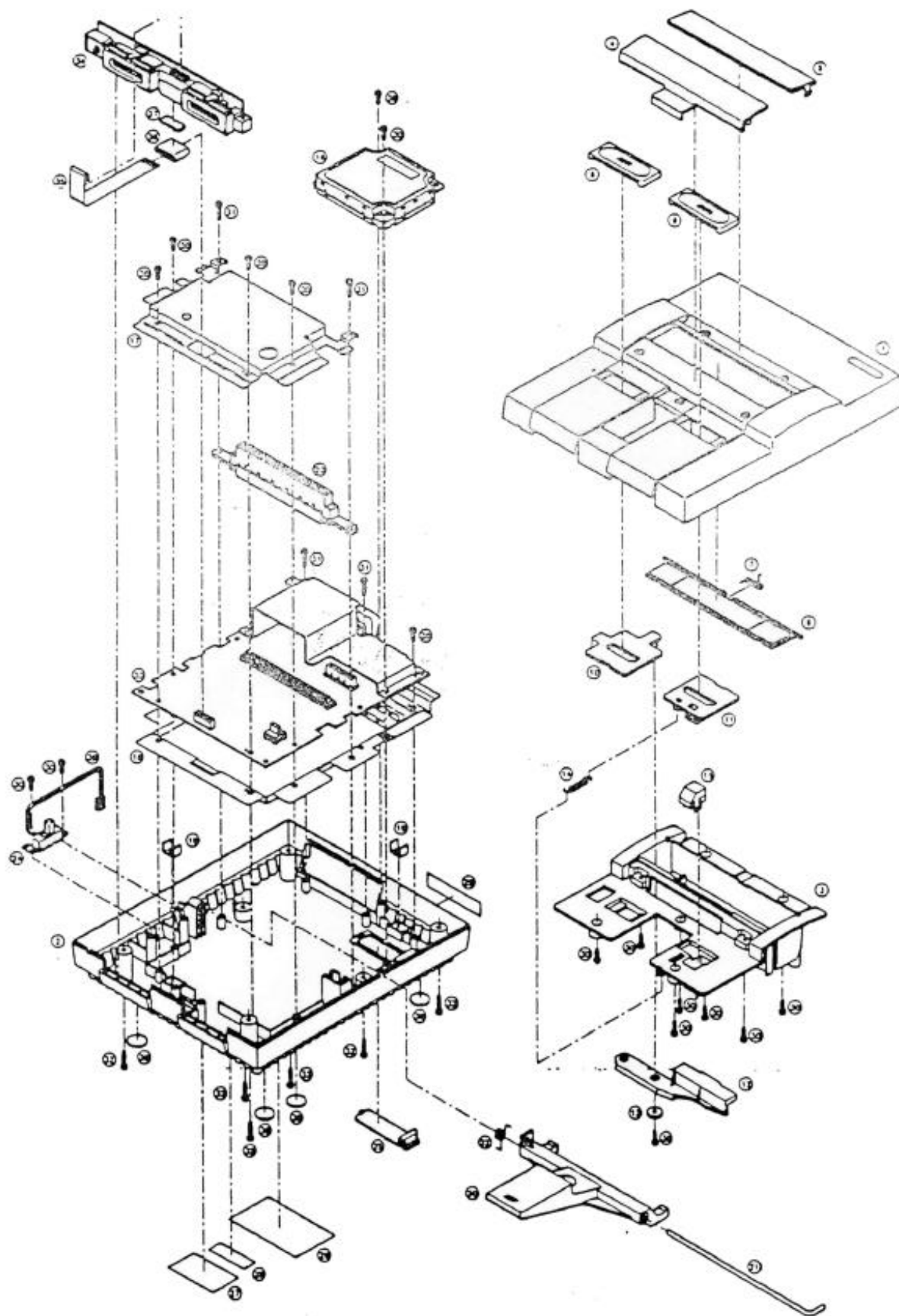
2.7 Display Section

PPU1 and PPU2 work together with two 32 KB VRAM SRAMs, an RGB encoder (BA6592F or similar), and output amplifiers. PPU1 primarily handles object (sprite) generation, rotation/scaling, and background address calculation. PPU2 handles color, windows, mosaics, color math, and final RGB generation. The RGB encoder produces composite video and S-Video.

2.8 Program Section

Contains the main CPU and 128 KB Work RAM. The CPU fetches and executes instructions from cartridge ROM, generates timing and control signals, manages DMA/HDMA transfers, performs arithmetic and logic operations, and responds to interrupts and reset.





**NOTE: The circled numbers correspond to item numbers in the following parts list.
(These circled numbers are not part numbers).**

ITEM	CODE	DESCRIPTION	QUANTITY
38	340.001.1	Rubber Feet	4
16	640.003.1	Sound Module	1
36	123.001.1	Ferrite Ring	1
17	321.001.1	Upper Shielding	1
18	321.004.1	Lower Shielding	1
23	240.001.1	62-pin Connector Upper	1
1	330.001.1	Upper Shell	1
2	330.005.1	Lower Shell	1
	240.012.1	12-pin connector	1
	240.013.1	28-pin connector	1
	240.018.1	21-pin FFC.S connector	1
	153.002.1	Integrated Circuit CPU RF5A22	1
	153.005.1	Integrated Circuit PPU1 RF5C77	1
	153.006.1	Integrated Circuit PPU2 RF5C78	1
	152.004.1	Integrated Circuit WRAM 1M	1
	154.001.1	SOP Integrated Circuit BA6592F RGB	1
	170.001.1	Axial Fuse 1.5A SSTRr	1
	135.002.1	Diode Array DAN202	5
	135.003.1	Diode Array DAP202	5
	140.001.1	Transistor 2SA1037	6
	140.003.1	Transistor 2SC2412	10
	140.002.1	Transistor 2SC4699 NP	2
	240.003.1	62-pin Connector Lower	1
	155.001.1	Integrated Circuit F411 CIC	1
	124.001.1	Power Filter ZJYS5012.2PT	1
	161.002.1	Ceramic Chip Oscillator 4 MHz	1
	160.002.1	Crystal Oscillator 21 MHz	
	150.001.1	Integrated Circuit 74HCU04 SOP	1
	154.002.1	Integrated Circuit LM 358 SOP	1
	152.003.1	Integrated Circuit 256K S.RAM 28P SOP	2

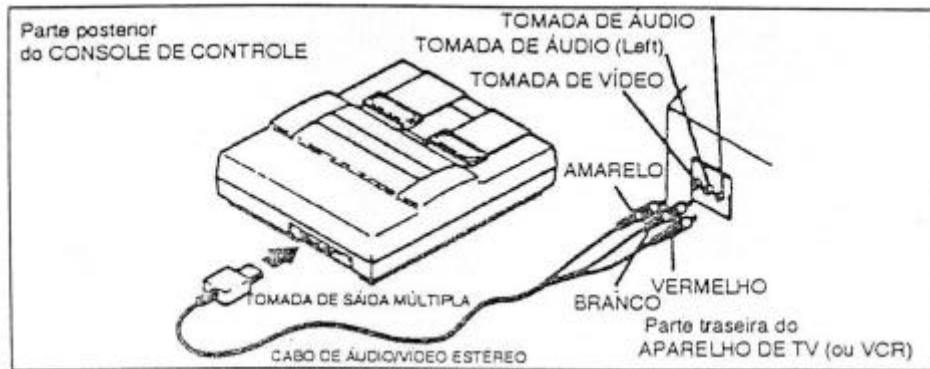
ITEM	CODE	DESCRIPTION	QUANTITY
	156.001.1	Integrated Circuit 7805	1
	320.001.1	Heatsink	1
	301.001.1	Insulating Washer	1
	300.016.1	Screw 3x10	1
	300.017.1	Screw 3x7	5
	641.001.1	RF Modulator. SNES	1
	240.011.1	Power Jack	1
	260.006.1	Reset Button	1
	132.003.1	Diode SK. RB110C	1
	155.002.1	Integrated Circuit PST529DMT Reset	1
	105.002.1	Varistor ERZ. CF2MK220	1
	120.002.1	Inductor Chip 22uH NL322522T.22	1
	110.003.1	Ceramic Chip Capacitor 0.1 uF/25V	6
	110.001.1	Ceramic Chip Capacitor 0.1 uF/50V	15
	110.002.1	Ceramic Chip Capacitor 0.047uF/25V	1
	110.006.1	Ceramic Chip Capacitor 10pF/50V	4
	110.013.1	Ceramic Chip Capacitor 47pF/50V	7
	110.014.1	Ceramic Chip Capacitor 56pF/50V	3
	110.015.1	Ceramic Chip Capacitor 82pF/50V	4
	110.010.1	Ceramic Chip Capacitor 220pF/50V	2
	110.011.1	Ceramic Chip Capacitor 300pF/50V	13
	110.004.1	Ceramic Chip Capacitor 1000pF/50V	1
	110.009.1	Ceramic Chip Capacitor 1 uF/16V	16
	112.003.1	Electrolytic Chip Capacitor 10uF/16V	3
	112.005.1	Electrolytic Chip Capacitor 33uF/25V	2
	112.004.1	Electrolytic Chip Capacitor 2.2uF/25V	1
	100.018.1	Resistor Chip 1/10W 39 OHM	3
	100.020.1	Resistor Chip 1/10W 56 OHM	1
	100.011.1	Resistor Chip 1/10W 100 OHM	13
	100.012.1	Resistor Chip 1/10W 120 OHM	3

ITEM	CODE	DESCRIPTION	QUANTITY
	100.013.1	Resistor Chip 1/10W 130 OHM	3
	100.014.1	Resistor Chip 1/10W 180 OHM	3
	100.015.1	Resistor Chip 1/10W 200 OHM	6
	100.016.1	Resistor Chip 1/10W 270 OHM	3
	100.017.1	Resistor Chip 1/10W 300 OHM	4
	100.019.1	Resistor Chip 1/10W 470 OHM	3
	100.023.1	Resistor Chip 1/10W 560 OHM	1
	100.007.1	Resistor Chip 1/10W 1K OHM	9
	100.003.1	Resistor Chip 1/10W 1.2K OHM	2
	100.004.1	Resistor Chip 1/10W 1.8K OHM	5
	100.009.1	Resistor Chip 1/10W 3.3K OHM	2
	100.006.1	Resistor Chip 1/10W 10K OHM	25
	100.008.1	Resistor Chip 1/10W 24K OHM	3
	100.010.1	Resistor Chip 1/10W 33K OHM	2
	100.005.1	Resistor Chip 1/10W 100K OHM	1
	300.018.1	Screw M3x8 w/washer	1
30	300.002.1	Screw M3x10 2N AN	17
31	300.001.1	Screw M3x14	4
32	300.003.1	Special Screw N76	6
25	331.001.1	Expansion Connector Cover	1
4	331.003.1	Rear Bottom Panel Case Part	1
5	331.002.1	Rear Top Panel Case Part	1
6	331.033.1	Cartridge Cover - NES	1
7	310.002.1	Cartridge Door Spring	1
8	333.002.1	Power Button	1
9	333.003.1	Reset Button	1
10	322.004.1	Power Button Support	1
11	322.005.1	Reset Button Support	1
19	322.001.1	PCI Support	1
34	240.002.1	Sub. Conj. Connector Module	1

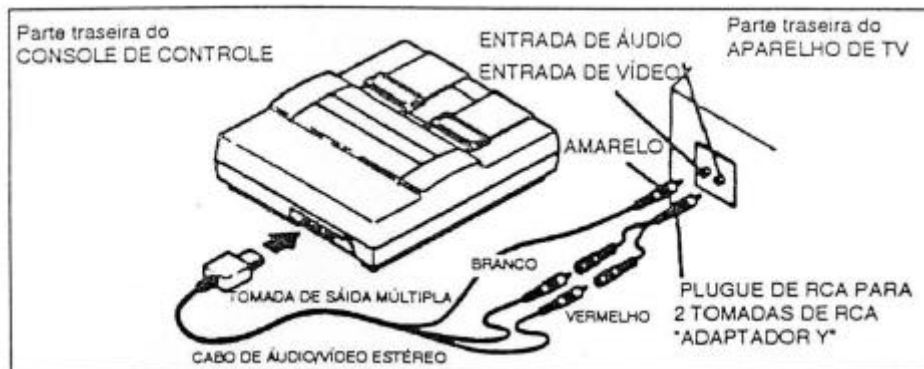
ITEM	CODE	DESCRIPTION	QUANTITY
24	260.001.1	Power Switch	1
20	333.001.1	Eject Button	1
22	310.001.1	Eject Button Spring	1
14	310.003.1	Reset Button Spring	1
21	242.001.1	Eject Button Lock Pin	1
37	341.001.1	Self-adhesive shim	1
39	210.001.1	Wiring Assembly for Power Switch	1
35	201.004.1	Parallel Cable	1
12	336.011.1	SNES Cartridge Lock (IMP)	1
13	334.003.1	SNES Cartridge Lock Holder (IMP)	1
15	333.008.1	Reset button extension	1
3	322.003.1	Chassis	1

CONNECTION

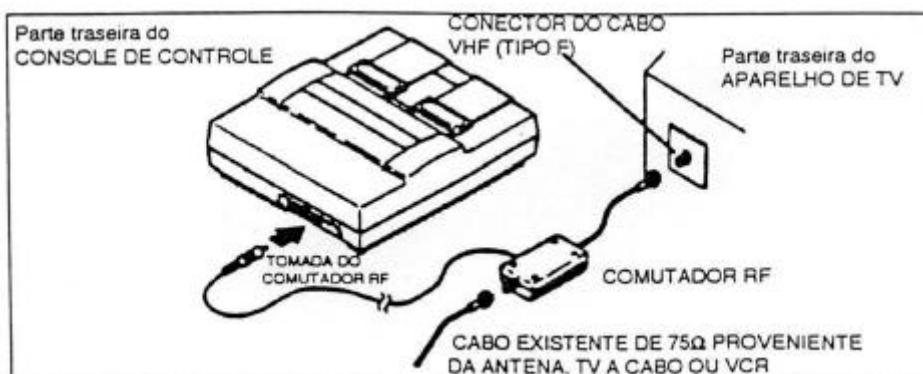
The Super Nintendo Entertainment System can be connected to a TV or VCR. There are several options for connection. If the TV (or VCR) has a left and right AUDIO IN and a VIDEO IN, the Super NES can be connected as shown below.



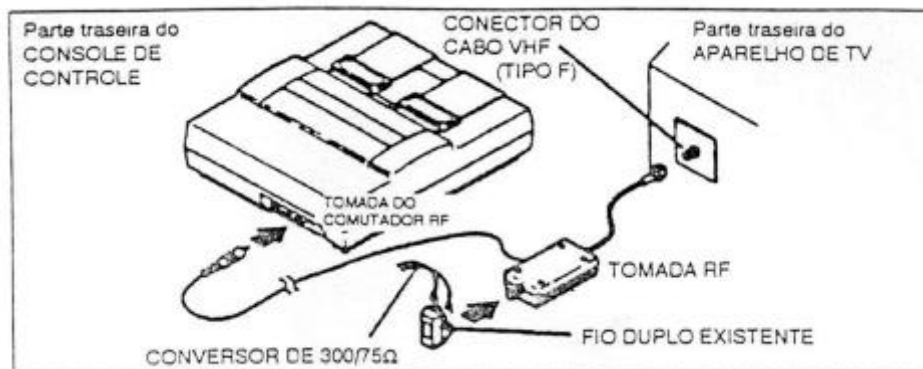
If the TV (or VCR) has an AUDIO IN and a VIDEO IN, a "Y-adapter" should be used. The Y-adapter can be purchased at an electronic parts store.



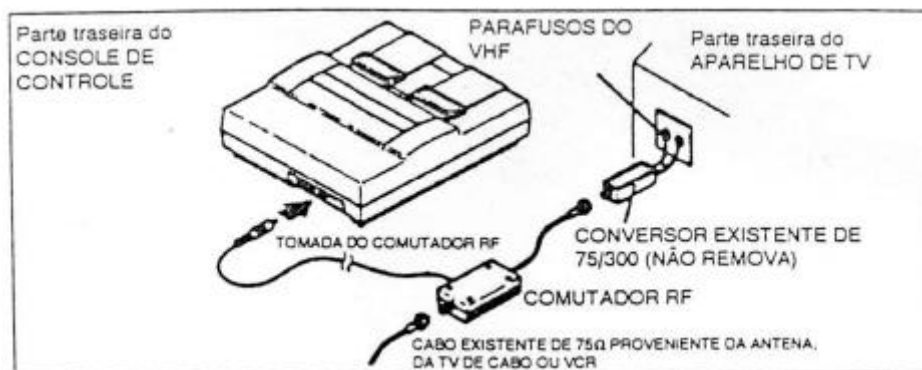
If the TV (or VCR) has a VHF (Type F) cable connector, connect the Super NES as shown below.



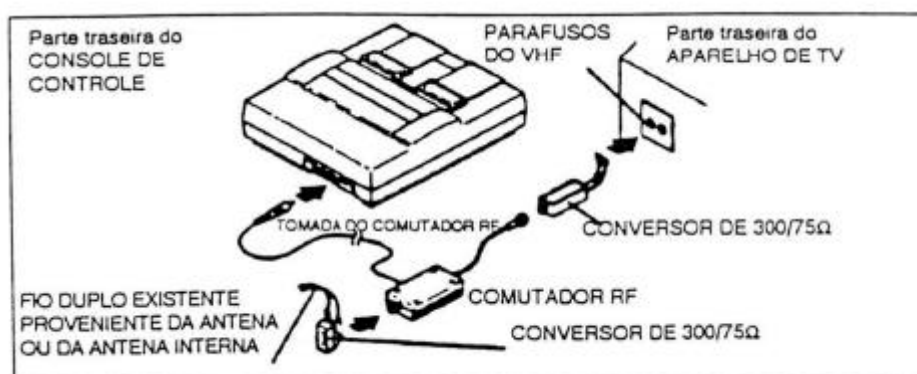
If the (TV or VCR) has a VHF cable connector that uses a 300/75Ω converter, connect the Super NES as shown below.



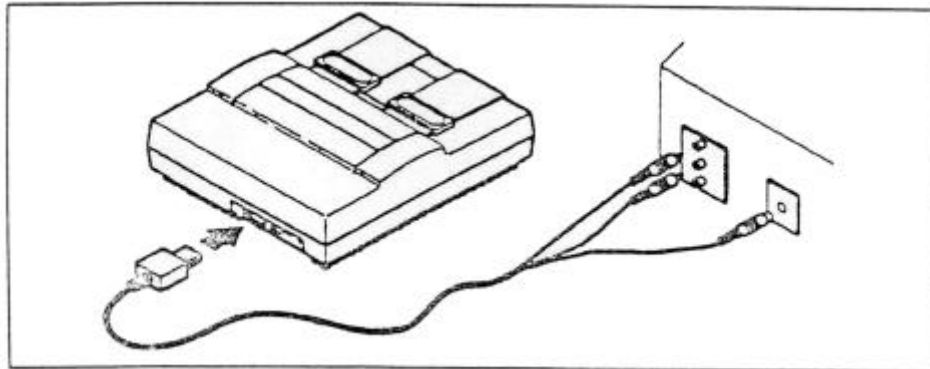
If the TV uses an F-type VHF connector and a 75/300Ω cable, use the connection method shown below.



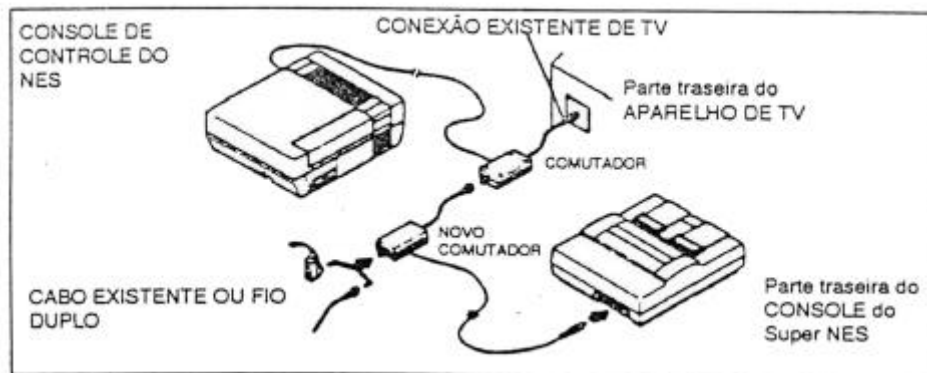
If the TV uses a screw-type VHF connector and dual wires, the 75/300Ω converter and a 300/75Ω converter will need to be connected as shown below.



If a TV, VCR or Stereo receiver accepts the S-Video input, an S-Video cable (part no. 22425) can be used to connect the Super NES as shown below. (NOTE: The S-Video cable is not included with the Super NES).



The Super NES can be connected along with the NES to a TV, monitor or VCR as shown below.



NOTE: In some cases, this connection may cause attenuation of channels higher than 13.

GENERAL DISASSEMBLY INSTRUCTIONS

Disassembling the Console

1. **Upper Cabinet:** Before opening, power the console on with the AC adapter unplugged to discharge capacitor C67 (this residual charge can damage components during servicing). Place the unit upside-down and remove the six security screws with a GameBit screwdriver. Turn the unit over and lift off the top shell.
2. **Sound Module:** Remove the two screws securing the sound module and lift it vertically.
3. **Cartridge Mechanism:** Lift the ejector bar, slide it to the right, and remove the entire mechanism. Note the position of the ejector spring for correct reassembly.
4. **Power Switch:** Remove its two screws, disconnect the 2-pin cable, and lift out the switch assembly.
5. **Controller Port PCB:** The front controller PCB is connected by an 11-way ribbon cable. Gently pull the cable free and remove the PCB.
6. **Upper Shield:** Remove the six screws holding the upper RF shield. Most of the main PCB is now visible. Remove the remaining three screws at the rear. Note that the longer chrome screws belong next to the 62-pin connector and near the output ports.
7. **62-Pin Connector & Main PCB:** Grasp the top of the 62-pin cartridge connector and pull firmly upward to free it. Lift the main PCB out of the lower shell.

Disassembly of the Upper Cabinet

1. Place the upper shell upside-down and remove the reset spring with needle-nose pliers.
2. Remove the seven visible screws. Note the location of any support washer on the cartridge lock plate.
3. Lift the cartridge lock plate vertically, then toward you, and remove it.
4. Remove the final screw and lift the cartridge socket out of the top shell.
5. Hold the upper shell and push the front door cover upward while pulling it toward you. Note the door spring position for reassembly.
6. With power and reset buttons in the OFF position, lift the logo panel vertically out of the shell.
7. Compress the four plastic tabs and push the power and reset buttons downward to remove them.

CONSOLE TEST PROCEDURE

Equipment Required: Stereo television/monitor, Super NES AC adapter, RF switch, stereo A/V cables, two Super NES controllers, Super Mario World (or equivalent known-good game), Super NES Burn-In / Test Cartridge.

Before inserting any cartridge, visually inspect the 62-pin connector for bent pins, corrosion, or contamination. Do not attempt to clean the 62-pin connectors — replace them if damaged.

1. Connect AC adapter, RF switch (or A/V cables), and both controllers.
2. Insert Super Mario World and power on. Select the correct RF channel (3 or 4) or switch to A/V input.
3. Start a two-player game and play briefly. Verify color, image stability, sound, and both controllers.
4. Test both RF channels and wiggle the RF connector to check for intermittent contact. Switch to A/V mode and verify stereo audio and clean composite/S-Video output.

NOTE: Before attempting any PCB-level repairs, re-test the unit with a known-good sound module and a known-good upper 62-pin connector.

TROUBLESHOOTING

Burn-In Test — Priority for Replacement of Defective Components

The burn-in test cartridge verifies system RAM and registers, CPU read/write functions, and returned values. When multiple errors are reported, consider all factors; the CPU exercises other LSIs through the bus.

Test	1st	2nd	3rd	4th
WRAM	CPU	WRAM		
DRAM	CPU	WRAM		
VRAM (LOW)	PPU1	CPU	SRAM (U5)	PPU2
VRAM (HIGH)	PPU1	CPU	SRAM (U4)	PPU2
DMA MEMORY	CPU			
OAM	PPU1	CPU		
CGRAM	PPU2	CPU		
MPY 8×8	CPU			
MPY 16×8	PPU1	CPU		
DIV 16/8	CPU			
DMA	CPU	PPU1	PPU2	
EXT LATCH	PPU1	CPU		
HV TIMER	CPU	PPU2	PPU1	
VRAM COUNT	PPU1	CPU		
V224/V239	CPU	PPU2	PPU1	
FIELD FLAG	PPU2	CPU	PPU1	
VH FLAG	PPU2	CPU	PPU1	
OBJ L OVER	PPU1	CPU		
CPU (CLOCK)	CPU			
APU	Sound Module	CPU		

Color Tone Defects

Color problems after PPU2 are usually caused by the RGB encoder, discrete amplifiers, or the 21 MHz clock. Identify which outputs are affected (composite, RGB, S-Video, RF).

Monochrome screen: Check 21.47727 MHz (NTSC) oscillation. If unstable, replace trimmer capacitor TC1, then the 21 MHz crystal (X1). If only RF is bad, replace the RF modulator. If both RF and composite are bad, check soldering around the RGB encoder (U7) and replace U7 then PPU2.

Wrong color (reddish/bluish): If RGB and S-Video are both affected, suspect PPU2 or the RGB amplifier transistors (Q3–Q9). If only composite is wrong while RGB is good, suspect the RGB encoder or color filters (C14, C16).

Image Defects — Recommended Replacement Sequence

Vertical lines (characters OK): PPU2 → PPU1 → CPU → VRAM

Vertical lines (characters also faulty): PPU1 → PPU2 → CPU → VRAM

Mosaic / defective characters: PPU1 → PPU2 → CPU → VRAM

Object priority / color priority faults: PPU1 → PPU2 → CPU

Screen jitter / missing portion of screen: PPU1 → PPU2 → CPU → VRAM

Window position shifted: CPU → PPU2 → PPU1

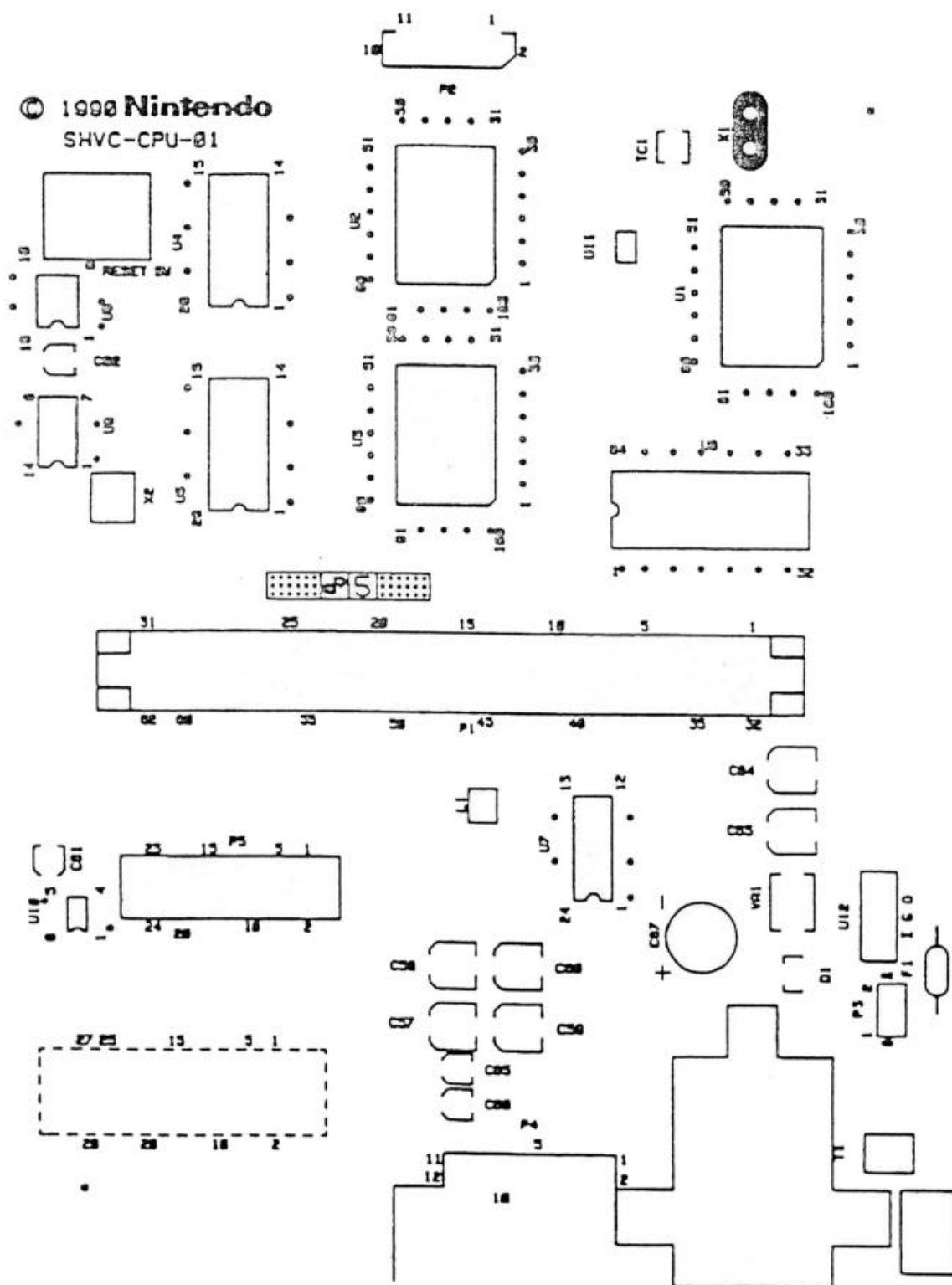
Specific cartridge only: CPU → PPU2 → PPU1

CONSOLE DIAGNOSTICS TABLE

Before any board-level work, fully inspect the PCB for damaged components, bad solder joints, corrosion, or missing parts. Do not attempt to clean the 62-pin connectors — replace them.

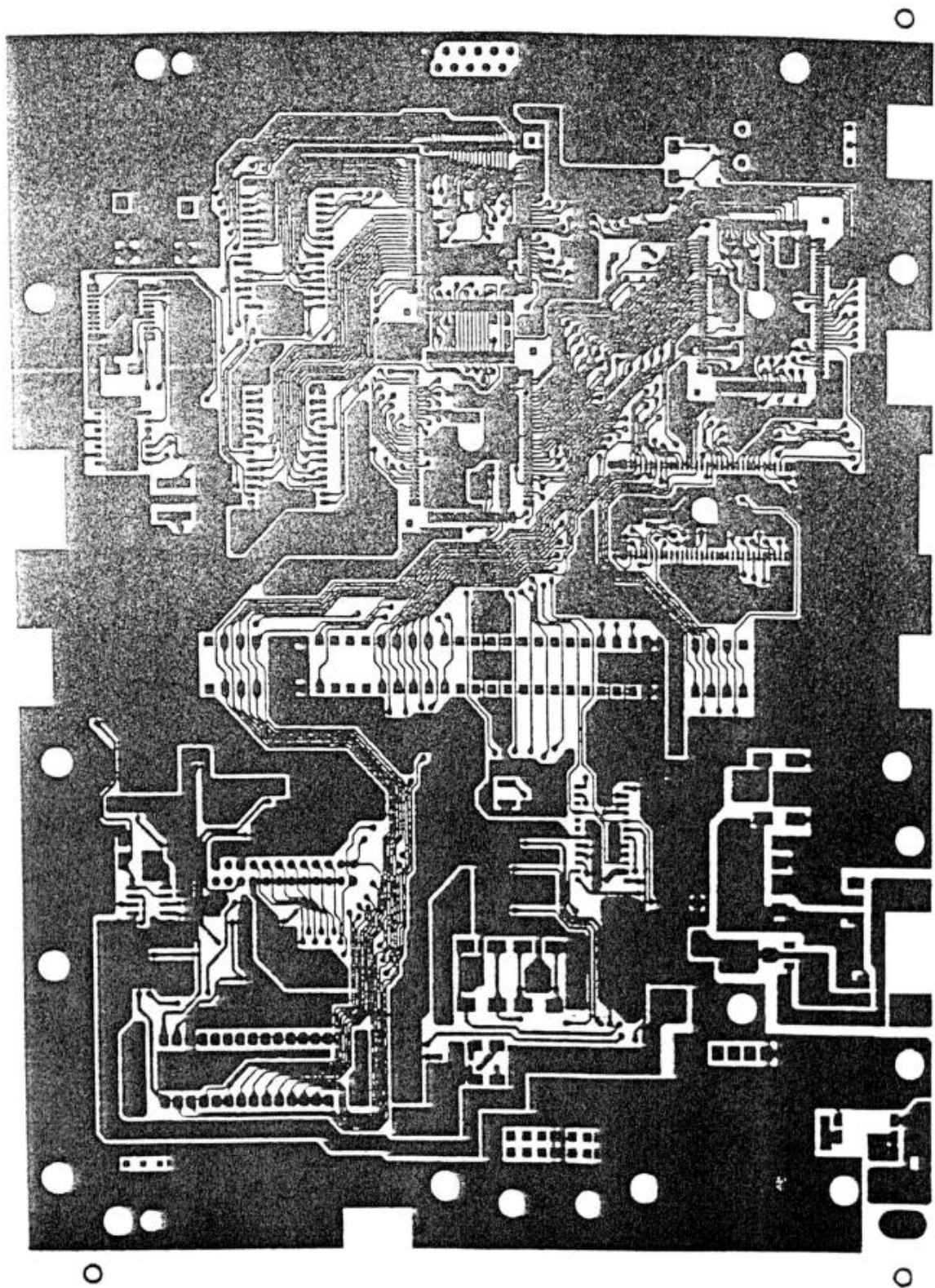
Malfunction	Probable Cause (priority order)
Solid color screen	Faulty 62-pin connector → Sound module → Poor soldering → CPU → PPU2 → WRAM → PPU1 → RGB encoder → 21 MHz oscillator → CIC
Distorted video	Faulty 62-pin connector → Poor soldering → PPU2 → CPU → PPU1 → VRAM SRAMs
Distorted or missing sound	Sound module → P5 connector → CPU → Audio amp (LM358) → Related transistors
No RESET	CIC → 4 MHz oscillator → 74HCU04
Weak or missing color	Defective TC1 (master clock frequency) → PPU2 → RGB encoder → 21 MHz oscillator
No power	1.5 A fuse → Diode → Line filter → 7805 regulator → Varistor → Related transistor → Power switch
Screen interference	RF modulator
Intermittent or missing controller operation	Front controller PCB → Ribbon cable → P2 connector → CPU
Frozen screen / erratic program operation	62-pin connector → Poor soldering → CPU → PPU2 → PPU1

© 1990 Nintendo
SHVC-CPU-01



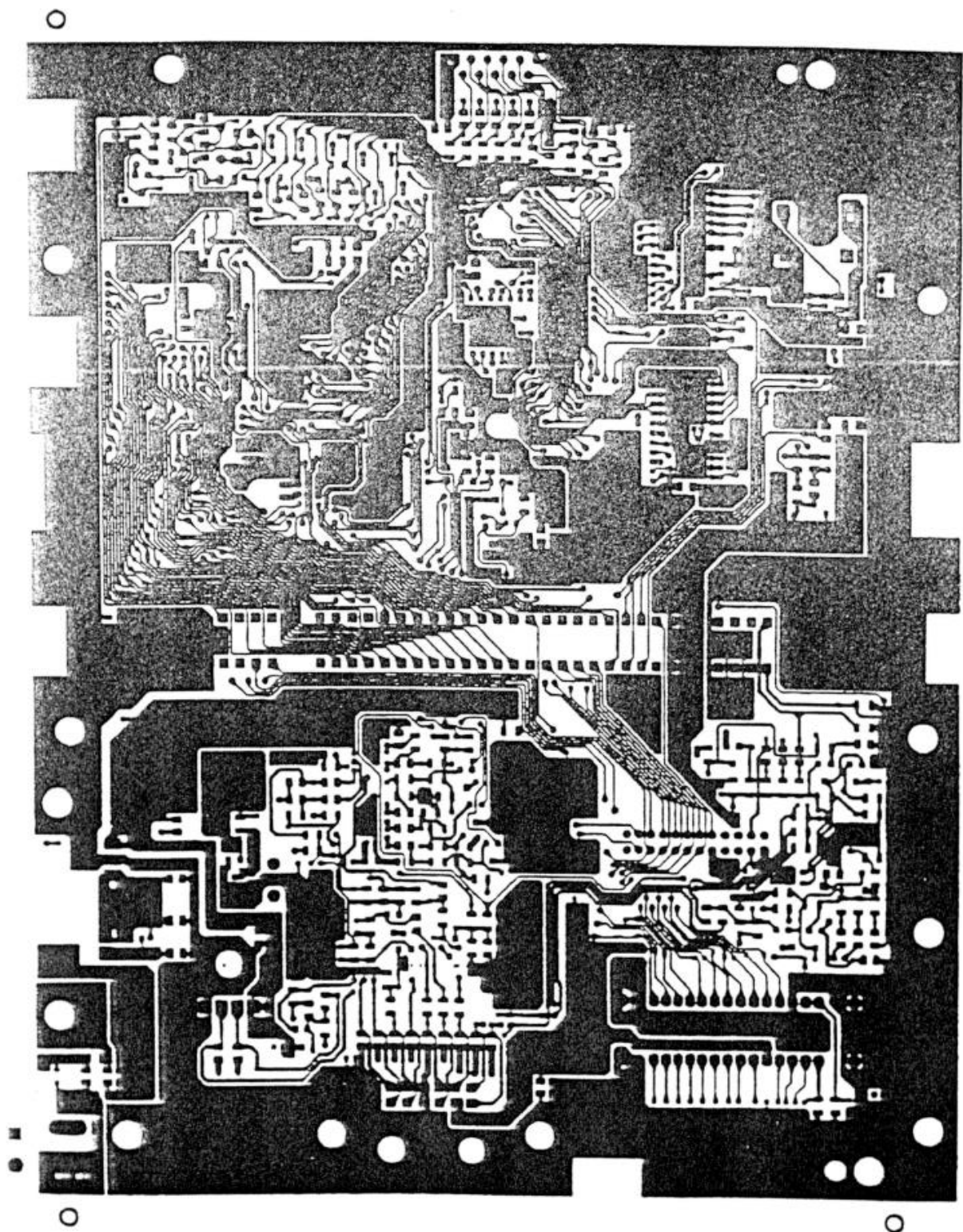
SHVC-CPU-01

COMPONENT LAYOUT (TOP SIDE)



SHVC-CPU-01
TRACK LAYOUT (TOP SIDE)

COMPONENT LAYOUT (UNDER SIDE)



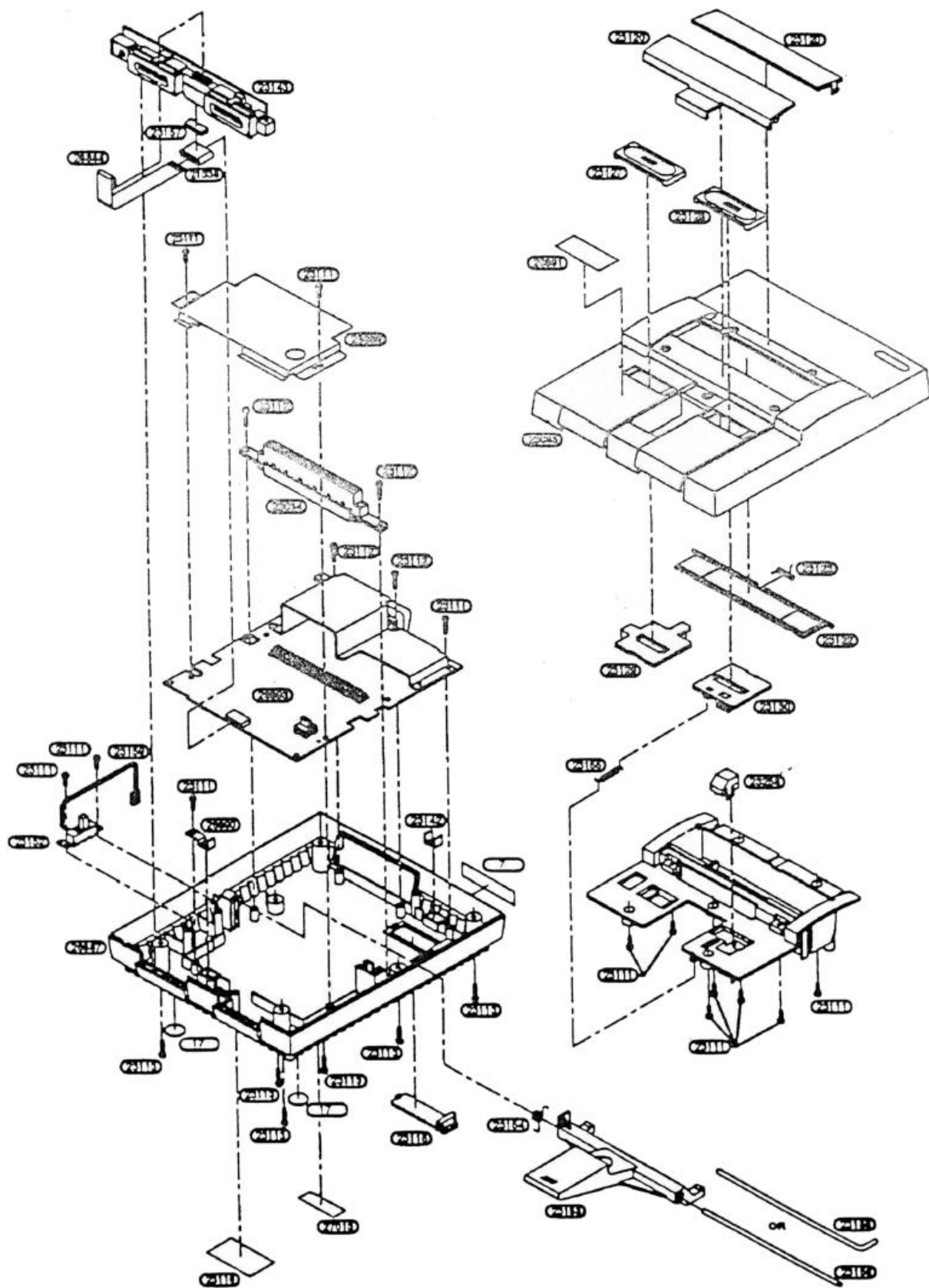
SHVC-CPU-01

TRACK LAYOUT (UNDER SIDE)

3. CONSOLE SUPER NES (GPM REVISION)

Later Super NES CPU boards use Ground Plane Method (GPM) construction. A ground plane layer is placed between the signal traces and the solder mask, eliminating the need for the bottom RF shield. Other changes include a shorter PCB, removal of capacitor C67 and the 4 MHz oscillator from their previous locations, integration of the sound circuitry onto the main board, conversion of TC1 to a through-hole component, and various discrete component count changes. The bottom shell and upper RF shield were modified accordingly. The mechanical cartridge lock bar was deleted and a warning label added to the top shell.

Functional theory of operation remains the same as the original SHVC board except for the System Timing and Reset sections (details were planned for a later update that was never issued). Use the same block diagram, connection, test, and diagnostic procedures as the original console.



ITEM	CODE	DESCRIPTION	QUANTITY
17	340.001.1	Rubber Feet	2
21338	123.001.1	Ferrite Ring	1
23084	240.001.1	62-pin Connector Upper	1
23085	330.001.1	Upper Case	1
23111	300.002.1	Screw M3x10 2N AN	16
23112	300.001.1	Screw M3x14	4
23113	300.003.1	Special Screw N76	6
23118	331.001.1	Expansion Connector Cover	1
23120	331.003.1	Rear Bottom Panel Chassis Part	1
23121	331.002.1	Rear Top Panel Chassis Part	1
23122	331.033.1	Cartridge Cover - NES	1
23123	310.002.1	Cartridge Door Spring	1
23127	333.002.1	Power Button	1
23128	333.003.1	Reset Button	1
23129	322.004.1	Power Button Support	1
23130	322.005.1	Reset Button Support	1
23142	322.001.1	PCI Support	1
23143	240.002.1	Sub. Conj. Connector Module	1
23152	260.001.1	Power Switch	1
23153	333.001.1	Eject Button	1
23154	310.001.1	Eject Button Spring	1
23155	310.003.1	Reset Spring	1
23156	242.001.1	Eject Button Lock Pin	1
23157	341.001.1	Self-adhesive shim	1
23159	210.001.1	Wiring Assembly for Power Switch	1
23258	333.008.1	Reset Button Trigger Extension	1
23259	322.003.1	Chassis	1
26887	330.002.1	Lower Case	1
26888	201.001.1	Parallel Cable (GPM)	1
26889	321.001.1	Upper Metal Shielding (GPM)	1

ITEM	CODE	DESCRIPTION	QUANTITY
26890	322.002.1	PCI Support (GPM)	1
	240.012.1	12-pin connector	1
	240.013.1	28-pin connector	1
	153.002.1	Integrated Circuit CPU RF5A22	1
	153.005.1	Integrated Circuit PPU1 RF5C77	1
	153.006.1	Integrated Circuit PPU2 RF5C78	1
	152.004.1	Integrated Circuit 1M WRAM	1
	154.001.1	Integrated Circuit BA6592F RGB SOP	1
	170.001.1	Fuse 1.5A SSTR Axial	1
	135.002.1	Diode Array DAN202	5
	135.003.1	Diode Array DAP202	5
	140.001.1	Transistor 2SA1037	6
	140.002.1	Transistor 2SC2412	10
	140.002.1	Transistor 2SC4699 NP	1
	240.003.1	62-pin Connector Lower	1
	155.001.1	Integrated Circuit CIC F411	1
	124.001.1	Line Filter ZJYS5012-2PT	1
	150.001.1	Integrated Circuit 74HCU04 SOP	1
	152.003.1	Integrated Circuit 256K S-RAM 28P SOP	1
	156.001.1	Integrated Circuit 7805	1
	320.001.1	Heatsink	1
	301.001.1	Insulating Washer	1
	300.016.1	Screw 3x10	1
	300.017.1	Screw 3x7	4
	641.001.1	RF Modulator - SNES	1
	240.011.1	Power Jack	1
	260.006.1	Reset Switch	1
	132.003.1	Diode SK-RB110C	1
	155.002.1	Integrated Circuit Reset PST529DMT	1
	120.002.1	Inductor Chip 22uH NL322522T-22	1

ITEM	CODE	DESCRIPTION	QUANTITY
	110.003.1	Ceramic Chip Capacitor 0.1 uF/25V	25
	110.001.1	Ceramic Chip Capacitor 0.01 uF/50V	3
	110.002.1	Ceramic Chip Capacitor 0.047uF/25V	1
	110.006.1	Ceramic Chip Capacitor 10pF/50V	4
	110.013.1	Ceramic Chip Capacitor 47pF/50V	7
	110.014.1	Ceramic Chip Capacitor 56pF/50V	1
	110.015.1	Ceramic Chip Capacitor 82pF/50V	4
	110.010.1	Ceramic Chip Capacitor 220pF/50V	4
	110.011.1	Ceramic Chip Capacitor 300pF/50V	13
	110.004.1	Ceramic Chip Capacitor 1000pF/50V	3
	110.009.1	Ceramic Chip Capacitor 1 uF/16V	16
	112.003.1	Ceramic Chip Capacitor 10uF/16V	3
	112.005.1	Ceramic Chip Capacitor 33uF/25V	2
	112.004.1	Ceramic Chip Capacitor 2.2uF/25V	1
	100.018.1	Resistor Chip 1/10W 39 OHM	4
	100.020.1	Resistor Chip 1/10W 56 OHM	1
	100.011.1	Resistor Chip 1/10W 100 OHM	13
	100.012.1	Resistor Chip 1/10W 120 OHM	3
	100.013.1	Resistor Chip 1/10W 130 OHM	3
	100.014.1	Resistor Chip 1/10W 180 OHM	3
	100.015.1	Resistor Chip 1/10W 200 OHM	5
	100.016.1	Resistor Chip 1/10W 270 OHM	3
	100.017.1	Resistor Chip 1/10W 300 OHM	4
	100.019.1	Resistor Chip 1 /10W 470 OHM	3
	100.007.1	Resistor Chip 1/10W 1K OHM	10
	100.003.1	Resistor Chip 1/10W 1.2K OHM	2
	100.004.1	Resistor Chip 1/10W 1.8K OHM	6
	100.009.1	Resistor Chip 1/10W 3.3K OHM	2
	100.006.1	Resistor Chip 1 /10W 10K OHM	25
	100.008.1	Resistor Chip 1/10W 24K OHM	3

ITEM	CODE	DESCRIPTION	QUANTITY
	100.010.1	Resistor Chip 1/10W 33K OHM	2
	100.005.1	Resistor Chip 1/10W 100K OHM	2
	240.017.1	11-Pin FCC-A Connector	1
	300.018.1	Screw M3x8 w/washer	1
	161.001.1	Ceramic Oscillator 24 MHZ	1
	154.003.1	Integrated Circuit LM324 SOP	1
	153.004.1	Integrated Circuit CXP 1100Q S-SMP	1
	153.003.1	Integrated Circuit CXD 1222Q S-DSP	1
	153.007.1	Integrated Circuit UPD6376	1
	330.018.1	Crystal Oscillator 21 MHZ	1
	152.002.1	Integrated Circuit 256K PSEUDO RAM SOP2	
	105.001.1	Varistor ERZ-C05DK220	1
	113.001.1	Capacitor Trimmer 20pF	1

4. CONTROLLER

Purpose

The Super NES Controller provides the primary player interface. It features an 8-way directional pad, Select and Start buttons, and six action buttons (A, B, X, Y, L, R) whose functions are defined by software.

Theory of Operation

Single-IC version: Uses a 14-bit parallel-to-serial shift register (56V520). Twelve parallel inputs are pulled low when the corresponding carbon contact pad is pressed. The console supplies VDD, VSS, Clock, and P/S (OUT0). An OUT0 pulse latches the button state; subsequent clock pulses shift the data out serially on the data line.

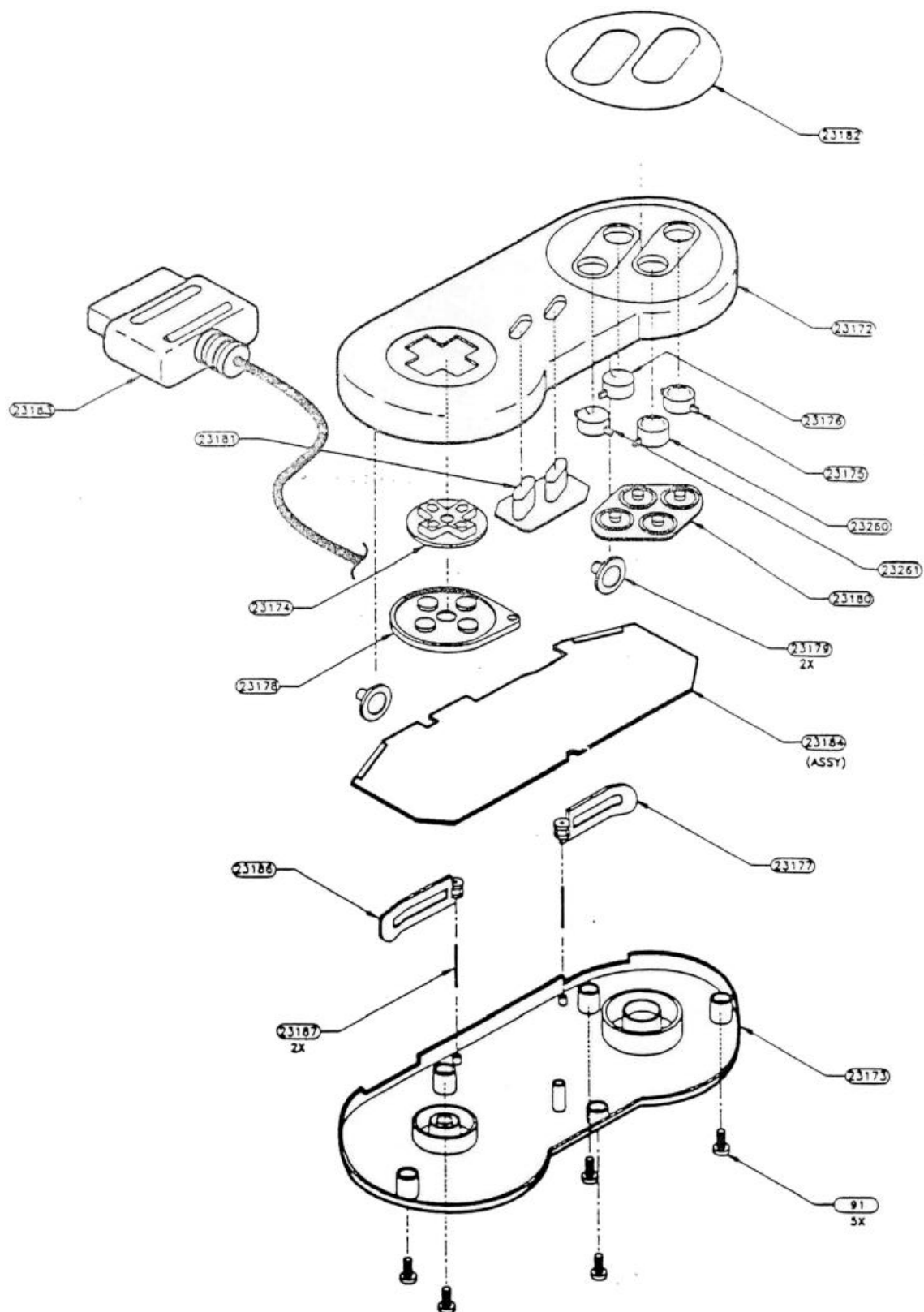
Dual-IC version: Uses two 8-bit parallel-to-serial shift registers (56W545). The first eight bits come from IC-A; the remaining four bits are daisy-chained from IC-B into the serial input of IC-A. Operation is otherwise identical.

Controller Test Procedure

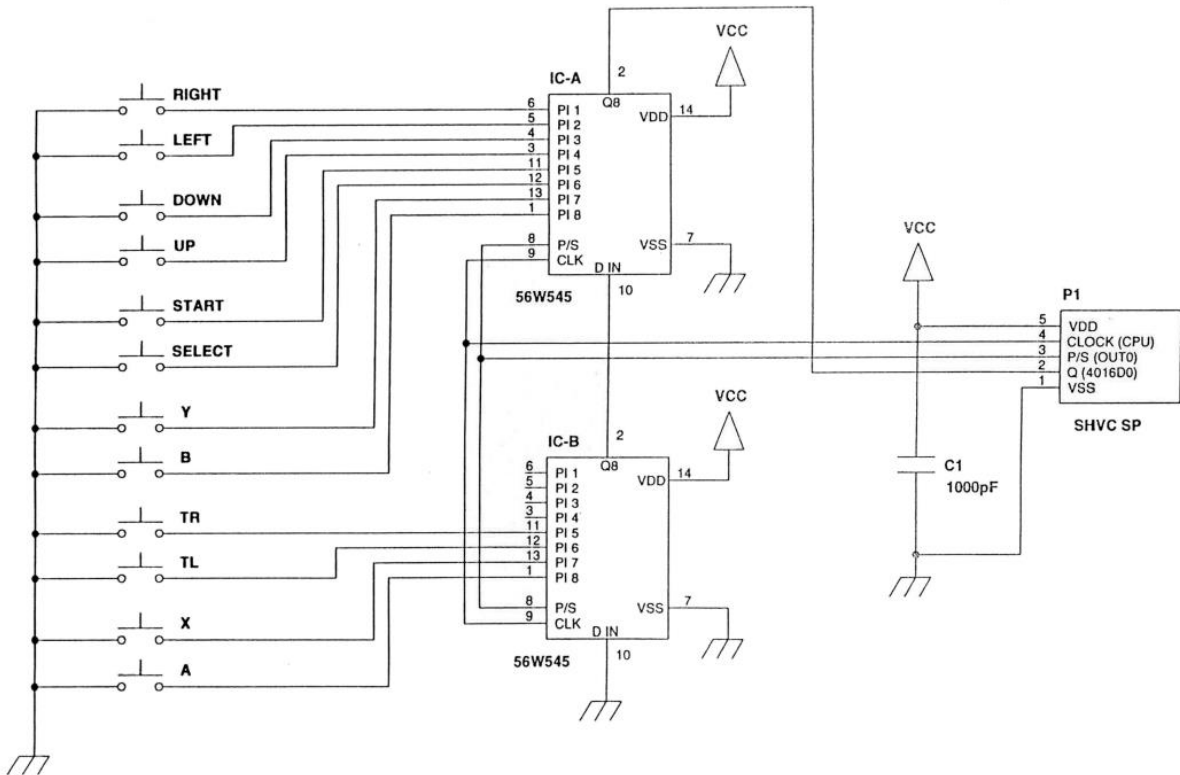
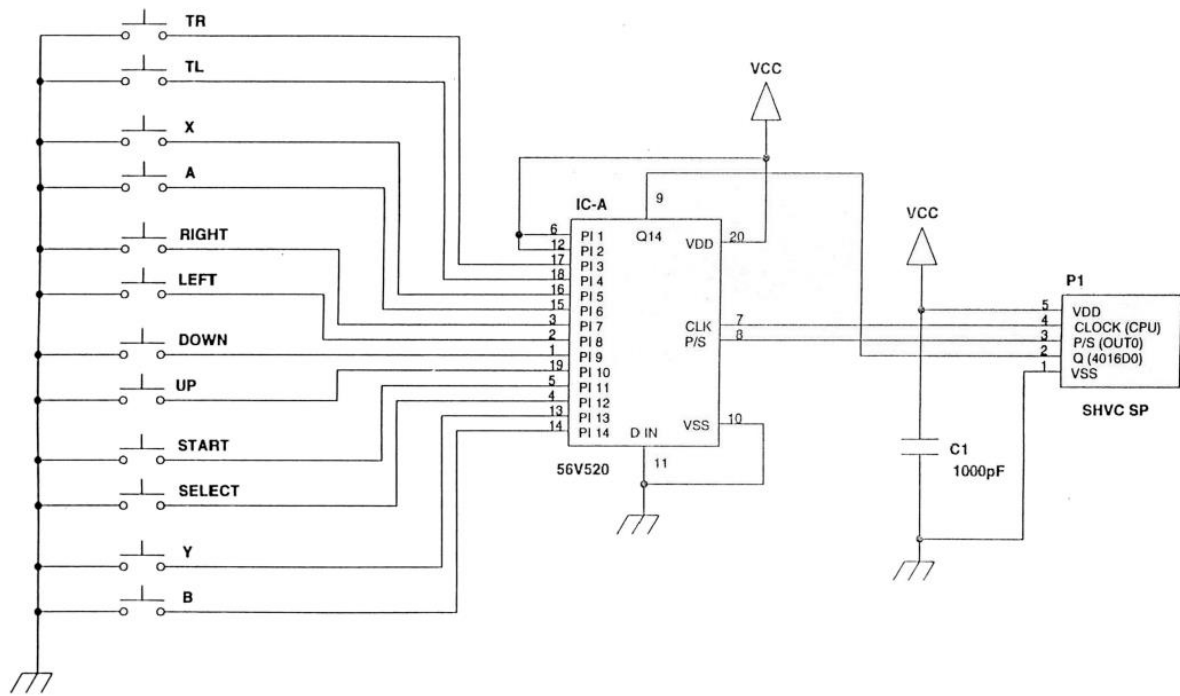
1. Connect console to a television and insert the Test Cartridge.
2. Plug the controller under test into a controller port and power on.
3. Select the Controller test. Press every button in the order indicated by the on-screen arrows. Each button should turn its screen region red while held and blue when released.
4. If no activity is detected within 10 seconds a fault is flagged.
5. Look for slow response, stuck buttons, or false triggers.
6. Disassemble and inspect contact pads, cable, and PCB. Replace worn contact membranes; do not attempt to clean them.

Controller Diagnostics

Symptom	Most Probable Causes
Slow response	Worn contact pads; dirty or damaged PCB contact surfaces
All functions fail	Faulty IC-A / IC-B
Multiple buttons register simultaneously	Faulty PCB or cable
Random or missing responses	Worn contacts; faulty IC; faulty cable or PCB



ITEM	CODE	DESCRIPTION	QUANTITY
91	300.019.1	Parafuso M2x8	5
23172	330.003.1	Upper Control Cabinet	1
23173	330.004.1	Lower Control Cabinet	1
23175	333.012.1	Button A	1
23176	333.010.1	Button X	1
23177	333.014.1	R Button	1
23178	342.001.1	Directional Contact Blanket	1
23179	342.004.1	Contact blanket Buttons R/L	2
23180	342.002.1	Button Contact Blanket A.B.X.Y	1
23181	342.003.1	Manta Select/Start	1
23182	540.134.1	Button Adhesive A,B,X,Y	1
23183	201.003.1	Controller Cable with 7-Pin Plug	1
23184	230.015.1	PCI Control Mounted	1
23186	333.015.1	L Button	1
23187	322.008.1	R/L Buttons Pin	2
23260	333.013.1	Button B	1
23261	333.011.1	Button Y	1



5. SUPER SCOPE

Purpose

The Super Scope is a wireless light-gun peripheral. The player aims at targets displayed on a CRT television. It consists of a battery-powered transmitter (the gun) and a receiver that plugs into controller port 2.

Theory of Operation

Transmitter: Powered by six AA batteries regulated to 5 V. Contains Trigger, Pause, Cursor, and Power/Turbo switches. An optical sensor detects CRT scan lines. The internal CPU encodes button state and tracking data and transmits them via infrared. The console's external latch captures the exact H/V coordinates of the electron beam at the moment of the trigger pulse. Optimal distance is approximately 3 meters (10 feet). Fluorescent lighting or screen reflections can interfere with operation.

Receiver: Draws power from the console (port 2). Contains an infrared receiver/amplifier, CPU, shift register, and tracking port. It decodes the IR signal and presents button and tracking data to the console on the standard controller data lines.

Super Scope Test Procedure

1. Place the receiver on top-center of the television and plug it into port 2.
2. Insert the Test Cartridge, power on, and select the Super Scope test.
3. Insert six AA batteries and set the power switch to ON (not TURBO).
4. From at least 1.5 m (preferably 3 m), aim at the center of the calibration target and press Fire.
5. Proceed through all test screens, checking single-shot, turbo, pause, and cursor functions.
6. Look for intermittent hits, shots that only register in the center, or missing turbo mode.

Super Scope Diagnostics — Transmitter

Symptom	Probable Causes
No power	Dead or weak batteries; bad battery contacts; faulty power switch; low voltage regulator; faulty G-AMP PCB
Intermittent or no response	Dirty or obstructed lens; faulty trigger switch; crushed internal cable; open diodes; faulty CPU PCB
No turbo operation	Faulty power/turbo switch; cold solder joints; faulty CPU PCB (or software does not use turbo)
Pause does not work	Faulty pause switch; cold solder; faulty CPU PCB
Cursor does not work	Faulty cursor switch; crushed internal cable; cold solder; faulty CPU PCB
Hits miss edges or bottom of screen	Aim calibration too close or too far; optical parallax of the eyepiece

Super Scope Diagnostics — Receiver

Symptom	Probable Causes
No power	Faulty controller cable; short or open on PCB; faulty receiver PCB
Shots hit only the center of the screen	Solder bridge on Q1; defective receiver PCB; receiver not centered on TV; dirty lens
Intermittent or missing operation	Faulty cable; cold solder joints; faulty receiver PCB

NOTE: The receiver must be plugged into port 2. Remove it when not in use; it can interfere with normal controller operation in some games.

24422	331.013.1	Left Side of Sight	1
24423	331.021.1	Mira Frontal	1
23111	300.002.1	Parafuso M3x10 2N AN	4
24425	331.014.1	Right Side	1
24426	331.015.1	Left main back	1
24427	331.016.1	Secondary right side	1
24428	331.017.1	Left Secondary Side	1
24429	339.001.1	Plastic line	1
24430	332.003.1	LED Display	1
24431	332.002.1	Lens	1
24432	331.022.1	Lens Protector	1
24433	333.007.1	Push button	1
24434	300.009.1	M2x10LH Steel Threaded Bolt	2
24435	220.001.1	Battery Contact	1
24436	540.007.1	Product Name Label	2
24440	339.002.1	Right stock	1
24441	339.003.1	Left stock	1
24443	331.023.1	Key holder	1
24444	333.004.1	Power Button	1
24445	335.005.1	Shutter button	1
24446	336.006.1	Pause Button	1
24448	202.001.1	Cabo tipo flat 4C-340L	1
ITEM	CODE	DESCRIPTION	QUANTITY
24449	202.002.1	Cabo tipo flat 4C-90L	1
24450	230.014.1	PCI Mounted Amplifier	1
24451	230.026.1	PCI CPU Mounted, Cursor	1
24452	260.004.1	Chave Power	1
24453	260.011.1	Chave Pause/Cursor	1

24454	260.003.1	Trip switch	1
24456	230.017.1	PCI Montado leds	1
24458	331.007.1	Battery Cover	1
152	340.002.1	Bores	4
23111	300.002.1	Parafuso M3x10 2N AN	4
24467	210.002.1	Conj. Entrance Socket with 7 Pins 1 Plug	1
24459	331.024.1	Tampa to Receptor	1
24460	331.019.1	Base do receptor	1
24461	332.004.1	Visor to receptor	1
24462	540.037.1	Receiver Label	1
24464	230.027.1	PCI Montado receptor	1

TEST PROCEDURE FOR SUPER SCOPE

Equipment then Needed:

1. Television Monitor

Console do Super NES

Controller do Super NES

Super NES AC adapter

Super NES Stereo A/V Cables or SNES RF Assembly Super NES Test Cartridge

Six AA Batteries

Procedure:

The Super Scope of the Super NES is sensitive to certain levels of light. When doing the test, make sure that there is no fluorescent light falling directly on the transmission tube or reflecting off the television screen. The brightness and contrast of your television also requires some adjustment.

1. Connect the A/V cables and the AC adapter. on the Console plug. Plug the Controller into the plug

of the n2 control socket 1, and the Super Scope Receiver assembly in the socket plug

6. MOUSE

Purpose

The Super NES Mouse (bundled with Mario Paint) can be used as a pointing device for drawing, painting, and music composition, or as a conventional controller depending on the software.

Theory of Operation

A custom IC processes the optical encoder signals generated by the ball and rollers together with the two button inputs. Movement and button data form a 32-bit packet that is shifted out on the 4016D0 line after each OUT0 pulse. Three tracking speeds are selectable by software via the OUT0 and clock signals. The mouse may be used in either controller port depending on the game.

Mouse Test Procedure

1. Plug the mouse into port 2 (port 1 is required for Mario Paint).
2. Insert the Test Cartridge and select the Mouse test.
3. Exercise the three test sections (electronics, buttons, cursor movement).
4. Look for jerky movement, stuck buttons, or no response.
5. Remove the ball and clean the rollers of dirt, hair, or fibers. Inspect the cable and shell for damage.

Mouse Diagnostics

Symptom	Probable Causes
Jerky or slow response	Tracking speed not set correctly by software; dirty ball or rollers; faulty cable or mechanical assembly
No response	Dirty ball/rollers; faulty cable; faulty PCB or mechanical assembly
Button test failed	Faulty top shell assembly or PCB

device for game play or as an instrument for creative expression (i.e., drawing, coloring, composing music).

Theory of Operation

The Super NES Mouse contains a special IC (U1) to process input information and create an output for the Console. Horizontal and vertical movement is determined by the Mouse Ball contacting the surface area and rotating the H/V rollers. The grooves in the disc, passing between the optical couplers, determine the speed and displacement in both horizontal and vertical directions. This motion data is combined with right/left key input data and velocity data, to form a 32-bit data sequence. This sequence is loaded into four 8-bit offset registers (internal with respect to U1). This information is then received by the Console on the 4016D0 line following each OUTO pulse (initialization signal) with CUPO pulses, gradually leaving the information. Three different speeds can be selected by the software via the OUTO and CUPO signals. The Super NES Mouse can be used on both ports, depending on the software programming.

(25992

25612)

SEE THE PARTS LIST ON THE FOLLOWING PAGE FOR AVAILABILITY

ITEM CODE DESCRIPTION QUANTITY

- 25468 336.012.1 Layer of the Mouse Sphere 1

- 25469 336.001.1 Mouse sphere 1

25470 336.002.1 Mouse Ball Protector 1

25612 201.006.1 Cabo do Mouse 1

25990 330.020.1 Upper Mouse Enclosure 1

25991 330.019.1 Mouse Bottom Enclosure 1

25993 300.024.1 Parafuso M2,6x8 Mouse Black 2

- 25994 300.023.1 Screw M2x6 Mouse Chrome 2

25995 230.173.1 PCI Mouse Mount 1

MOUSE TEST PROCEDURE

Equipment Required:

1. Television Monitor

Super NES Controller Console Super NES Super NES AC Adapter

7. CARTRIDGE

Purpose

The game cartridge supplies the program ROM, character data, optional battery-backed work RAM, the CIC security chip, and (on some titles) a DSP-1 or Super FX co-processor.

Theory of Operation

The CIC on the cartridge continuously authenticates with the CIC on the console. If communication fails, the console CIC holds the system in reset. Battery-backed SRAM (when present) retains save data after power is removed. The DSP-1 is a fixed-point math co-processor. The Super FX is a high-speed RISC co-processor that can perform 3-D polygon rendering, rotation/scaling, texture mapping, and lighting while running in parallel with the main CPU.

Game Cartridge Test Procedure

1. Clean the edge connector thoroughly.
2. If the cartridge contains a battery, measure its voltage (should be 2.7–3.2 V).
3. Insert into a known-good console and verify the title screen and basic gameplay.
4. If the cartridge has a save feature, create a save, power off for 3–5 minutes, and verify that the data is retained. Clear the save data before returning the cartridge to the customer.

Cartridge Diagnostics

Symptom	Most Probable Causes (priority order)
Blank / solid color screen	Faulty program ROM → CIC → 74LS139 → DSP-1 → Super FX → MAD-1 → PCB
Garbled video	Dirty or corroded edge connector → Super FX → Program ROM → DSP-1 → Work RAM → PCB
Frozen / erratic operation	Dirty edge connector → 74LS139 → DSP-1 → Super FX → Work RAM → PCB
Scrolling / layer problems	Program ROM or Work RAM
Save data lost	Dead battery or diodes → Faulty work RAM → 74LS139

Shortens processing time and prevents games from slowing down. In addition, Super FX has dedicated graphics functions that can quickly trace the images needed for: 3-D polygonal animation, object rotation and escalation, texture mapping, and light source shading. These elements are combined to create realistic simulation experiences.

Super NES Technical Manual

Cartridge

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(62P PCB)

PCB (POR GAME)

(62P PCB)

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ITEM CODE DESCRIPTION QUANTITY

7879 300.010.1 Gabinete LH M2x5.9 2

22393 331.009.1 Front Cabinet Cartridge 1

27243 331.026.1 Gabinete frontal PCB 62p 1

22940 331.018.1 Cartridge Rear Chassis 1

27244 331.027.1 62p 1 PCB Rear Enclosure 1

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SHVC-1C0N5S-01

Super NES Cartridge Technical Manual

GAME CARTRIDGE TEST PROCEDURE

Equipment then Needed:

1. Television Monitor

Console do Super NES

Super NES AC adapter

Super NES Stereo A/V Cables or SNES RF Assembly

8. NOTES ON HANDLING ICs

The custom ICs used in the Super NES are MOS devices and are extremely sensitive to static electricity and improper grounding of tools.

- A.** Always wear a grounded anti-static wrist strap that includes several megohms of series resistance. Avoid synthetic clothing and gloves that generate static. Prefer cotton clothing.
- B.** Before installation, store ICs on a grounded conductive surface or anti-static foam.
- C.** Use only a properly grounded soldering iron. Although the ICs contain protection diodes, sufficient voltage will destroy the diodes and the gate oxide.
- D.** Avoid touching surface-mount packages with bare fingers. Skin oils can be absorbed by the porous package material and expand when heated, cracking the package. Use finger cots or latex gloves. Do not reuse an IC that has already been desoldered.
- E.** Verify correct orientation before soldering. Reverse insertion will destroy the chip.
- F.** Disconnect power before inserting or removing any IC.
- G.** Transport ICs in anti-static trays or conductive foam so that all pins are at the same potential.
- H.** Avoid mechanical shock.
- I.** Although storage ratings may be $-20\text{ }^{\circ}\text{C}$ to $+70\text{ }^{\circ}\text{C}$, keep ICs near ambient temperature and away from high humidity.
- J.** Do not use a multimeter on resistance ranges that apply high test voltages to the pins. If resistance measurement is unavoidable, use the lowest range and proceed with extreme caution.
- K.** Keep soldering time as short as possible to minimize thermal shock to the package.

9. GLOSSARY OF TERMS

Color Manager RAM (CGRAM)

Internal RAM inside PPU2 that holds the 256-entry color palette. Loaded by DMA and addressed via the CGADD register.

Direct Memory Access (DMA)

Hardware-controlled block transfer that temporarily takes the bus from the CPU. Normally performed during horizontal or vertical blank to avoid visible artifacts.

Mosaic

Special effect that multiplies the size of pixels, reducing effective resolution for a blocky appearance.

Object Attribute Memory (OAM)

Internal RAM inside PPU1 that stores sprite attributes (tile number, palette, X/Y position, priority, size, flip flags).

Pulse Code Modulation (PCM)

Digital audio encoding technique used by the Super NES sound system.

RGB

Separate analog Red, Green, and Blue video signals intended for high-quality monitors.

Shift Register

Digital circuit that moves data one bit position on each clock pulse; used for serializing controller button data.

Video RAM (VRAM)

The 64 KB of static RAM used by the PPUs to store background maps, tile data, and sprite graphics.

Window

A hardware-defined rectangular region of the screen that can be used for clipping, color math, or special effects.

Work RAM

The 128 KB of general-purpose RAM used by the CPU for variables, sprite tables, scores, timers, and other runtime data.

10. ABBREVIATIONS AND ACRONYMS

ALE	Address Latch Enable	INT / IRQ	Interrupt / Interrupt Request
BE	Bus Enable	LSI	Large-Scale Integration
CA	CPU Address line	NMI	Non-Maskable Interrupt
CD	Control Data / Console	OE	Output Enable
CE	Chip Enable	OSC	Oscillator
CHR	Character (tile) data	PA	PPU Address line
CIC	Checking Integrated Circuit (lockout chip)	PBR	Program Bank Register
CONN	Connector	PCM	Pulse Code Modulation
CPU	Central Processing Unit	PPU	Picture Processing Unit
CTLR	Controller	PRG	Program
CS	Chip Select	RD	Read
CLK	Clock	RS	Register Select
DBR	Data Bank Register	RST	Reset
DMA	Direct Memory Access	SMP	Sound Microprocessor (S-SMP)
DSP	Digital Signal Processor	S-VIDEO	Separate luminance/chrominance video
EXP	Expansion	VDA / VPA	Valid Data Address / Valid Program Address
GPM	Ground Plane Method	WE / WR	Write Enable / Write
HSG	Housing	XSTR	Transistor

End of Super NES Technical Manual — Clean English Edition

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